

- 2 A salvage crane with an output power of 522 kW is lifting a fully submerged cube crate with sides of length 2.0 m out of the sea at a constant speed of 3.0 m s^{-1} as shown in Fig. 2.1.

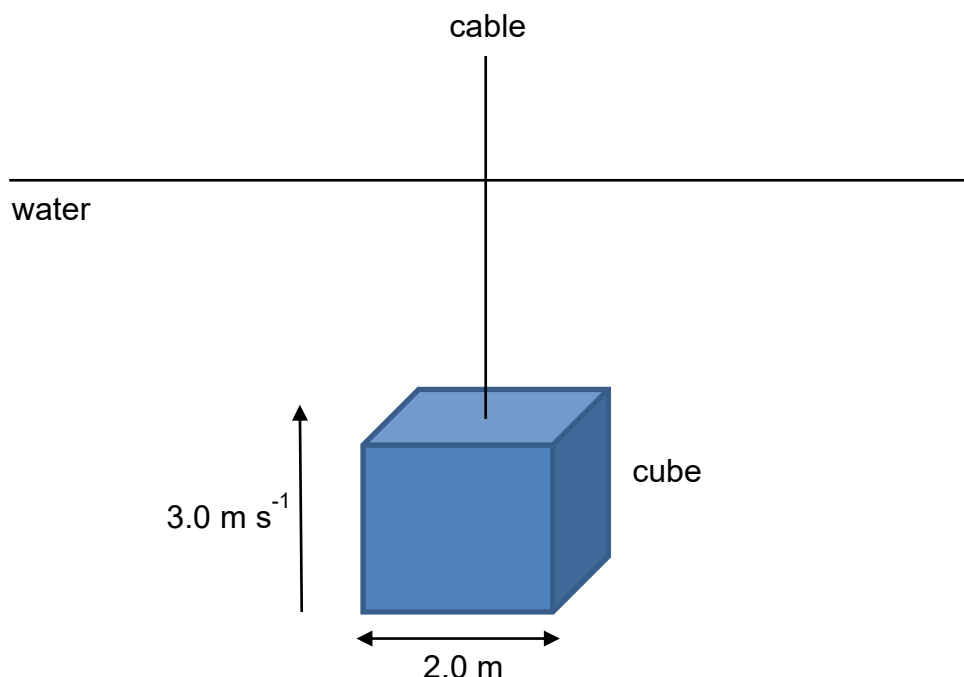


Fig. 2.1

The average density of the cube is 3060 kg m^{-3} and the density of the sea water is 1020 kg m^{-3} . The drag force of the water acting on the crate is not negligible.

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| (a) | (i) | State Archimedes' Principle. | |
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| | | | [1] |

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| | (i) | Solution:
Archimedes' Principle states that the upthrust acting on an object is equal in magnitude and opposite in direction to the weight of the fluid it displaces. | 1 |
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| | (ii) | Show that the upthrust acting on the crate is 80 kN. | |
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[1]

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| | (ii) | Solution:
Upthrust $= V \rho g = (2.0)^3 (1020)(9.81)$
$= 80049.6 \text{ N}$
$= 80 \text{ kN (shown)}$ | 1 |
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			[3]
	(c)	Solution: Upon leaving the surface of the water, the upthrust and drag force acting on the crate will be reduced to zero. Since the weight of the crate is larger than the force exerted by the crane, the direction of the net force acting on the crate will be downwards. As the direction of the net force is in the opposite direction of the displacement of the crate, net negative work will be done on the crate. The crate's speed will decrease. As the speed of the crate decreases, the force exerted by the crane will increase since the force is inversely proportional to the speed of crate. This leads to an increase in net force exerted on the crate.	1 1 1

3	A satellite of mass 2400 kg is projected from the Earth's surface and placed in a	
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